higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION – BOILER MAKING
NQF LEVEL 3

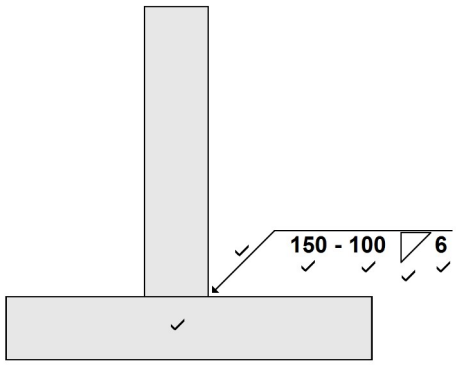
11 MARCH 2019

This marking guideline consists of 8 pages.

Copyright reserved

QUESTION 1

1.1



1 Mark – Welding joint, 1 Mark – Welding symbol, 1 Mark – 150 mm Pitch,
1 Mark – 100 mm Weld length, 1 Mark – Fillet weld symbol (Arrow side),
1 Mark – 6 mm Size of weld (6 × 1) (6)

1.2 FIGURE 2.1: Trammel✓ – To draw very large arcs, or very large circles.✓
To scribe circles where it is impractical to use dividers.

FIGURE 2.2: Prick punch✓ – To mark scribed lines permanently, or to punch specific points. ✓To act as guide when drilling.
(2 × 2) (4)

1.3 Center head✓ – Determining the center of a round object.✓/locate the center of an object. (2)

1.4
- Check all overall lengths.
- Examine each structural section like gussets.
- Verify sizes with drawing.
- Check drilling and cutting sizes.
- Make sure correct profiles have been used.
- Check marks or numbers of components.
- Check weld for defects. (Any 4 × 1) (4)

1.5
- Welding process requirements
- Base metal
- Base metal thickness
- Filler metal
- Weld type
- Welding position
- Shielding gas
- Pre-heat conditions.
- Post-weld heat treatment
- Welding progression
- Backing information (Any 4 × 1) (4)

[20]

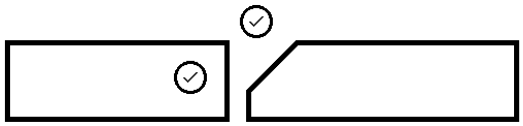
QUESTION 2

2.1
- Only use safe tools
- Use the correct tool for the job
- Use the tool correctly (3 × 1) (3)

2.2
- Irregular wire feed
- Porosity
- Unstable arc
- Difficulty of arc starting
- Burn back
- Metal hardening
- Reduction in metal fatigue strength
- Poor appearance
- Cannot weld in wind (Any 4 × 1) (4)

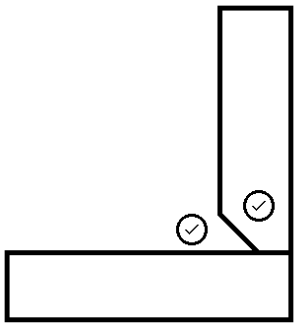
2.3
- All equipment must be cleaned after use
- Cables must be rolled up
- Repairs must be done immediately
- Regular maintenance checks must be done
- Ensure cylinders are closed
- All tools must be stored correctly (Any 2 × 1) (2)

2.4 2.4.1



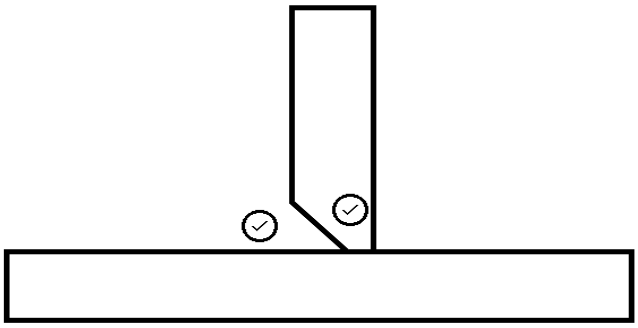
Single – V groove weld – Butt joint
(1 Mark – Single V groove, 1 Mark – Butt joint)

2.4.2



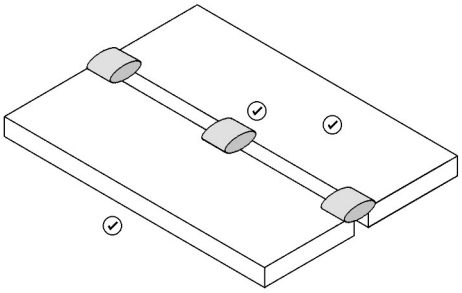
Single Bevel inside groove weld – Corner joint
(1 Mark – Single Bevel inside groove weld, 1 Mark – Corner joint)

2.4.3



Single Bevel groove weld – T joint
(1 Mark – Single Bevel groove weld, 1 Mark – T joint)
(3 × 2) (6)

2.5



(1 Mark – Isometric sketch, 1 Mark – Butt joint, 1 Mark – Tack welds)
(Any similar sketch to be accepted) (3 × 1) (3)

2.6
- Decrease the current
- Shorten the arc length
- Change the torch angle (3 × 1) (3)

2.7 Porosity blow holes in a weld✓ when gas trapped inside✓ a weld. (2 × 1) (2)

2.8
- Keep leads clear from walkways
- Keep equipment clear from water
- Ensure correct connections and earth
- Do not stand in water while welding
- Turn machines off after operation
- Check connections regularly (Any 4 × 1) (4)

2.9
- Copper
- Zinc (2 × 1) (2)

2.10
- Aluminum
- Copper
- Tin
- Zinc
- Antimony
- Lead (Any one answer) (1)

[30]

QUESTION 3

3.1

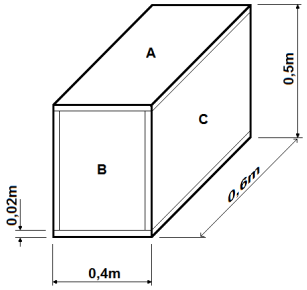


FIGURE 4

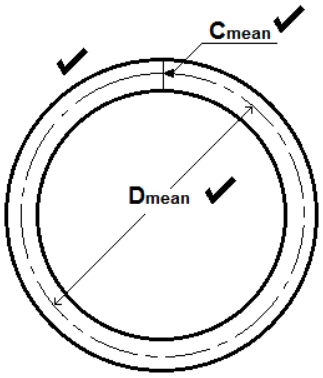
A_A = l × b ✓
= 0,600 × 0,400 ✓
= 0,240 m^2 → ✓

A_B = l × b ✓
= 0,460 × 0,360 ✓ (0,02 mm subtracted from four sides)
= 0,166m^2 → ✓

A_C = l × b ✓
= 0,600 × 0,460 ✓ (0,02 mm subtracted from top and bottom)
= 0,276 m^2 → ✓

A_{TOT} = (2 × A_A) + (2 × A_B) + (2 × A_C) ✓
= 0,480 + 0,332 + 0,552 ✓
= 1,364 m^2 → ✓ (3 × 4) (12)

3.2



1 Mark – Sketch, 1 Mark – C_{mean}, 1 Mark – D_{mean}) (3 × 1) (3)

3.3
- Operators can walk behind the load, having the best visibility.
- Operators can carry out with other ground tasks while waiting for the next hoisting job.
- Operators are free to move around enabling them to handle the loads faster. (3 × 1) (3)

3.4 3.4.1 Lower

3.4.2 Derricking jib down

(2 × 1) (2)

3.5
- Plate roll machine
- Slip roll machine (2)

3.6
- Provide for the health and safety of workers.
- Provide for the health and safety of people in connection with the workers.
- Establish an advisory council for OH&S.
- Provide for matters related to the OH&S act. (Any 3 × 1) (3)

[25]

QUESTION 4

4.1 *Tolerance* means it is the difference between✓ the minimum and maximum✓ size allowed when manufacturing the fabricated product. (2)

4.2 Allowance refers to the need to allow✓ for material thicknesses✓ during bending. Or it is the difference in size between two components or metal plates which are joined together. (2)

4.3

FRONT VIEW	TOP VIEW
OB	AB
OD	BC
	CD
	DA

(One mark for each identification) (6 × 1) (6)

4.4

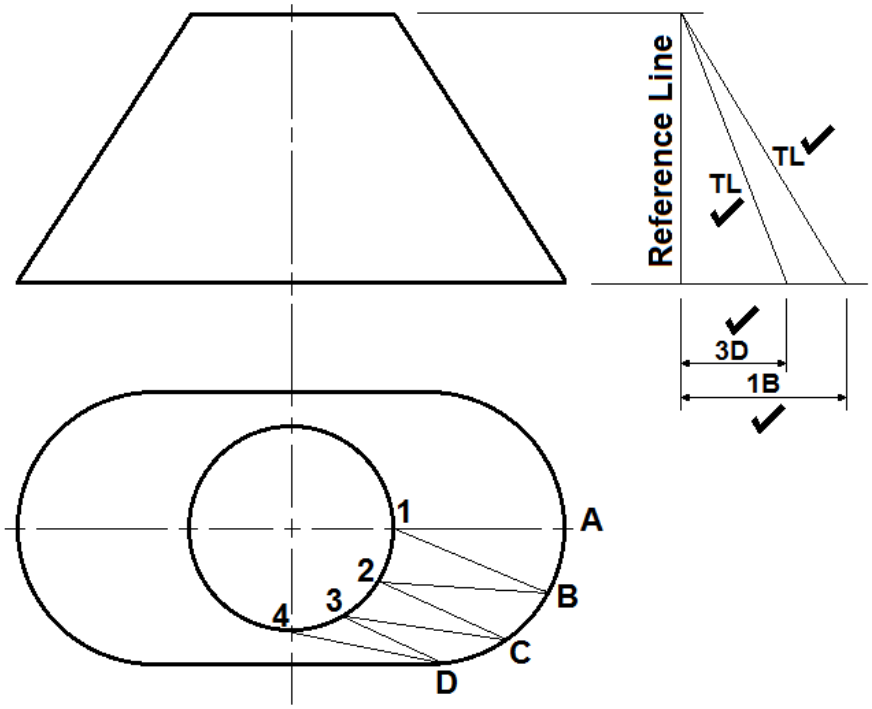


FIGURE 7
(1 Mark – Scale size 1B, 1 Mark – Scale size 3D)
(1 Mark – TL 1B, 1 Mark – TL 3D)
(Marks only be allocated for methods used and correct scale sizes) (4 × 1) (4)

4.5
- Wing
- Folding blade
- Handle
- Set screws
- Gauge adjustment screw
- Graduated scale
- Locking Screw
- Wedge adjusting screw
- Wedge lock nut
- 45° Stopper
- Adjustable collar
- 90° Stopper
- Frame (Any 2 × 1) (2)

4.6
- Ensure a clean area around the machine.
- Take note of safety procedures and signs.
- Wear PPE.
- Keep hands free from moving parts.
- Keep working area clean from any obstructions. (Any 2 × 1) (2)

4.7 Passivating (1)

4.8
- Poor penetration
- Undercut
- Welding slag
- Pit holds
- Porosity
- Slag inclusions
- Incomplete fusion
- Cracks (Any 4 × 1) (4)

4.9
- When stainless steel was heated and a coloured oxide layer was formed.
- When stainless steel was stored in an environment where it was subjected to contamination. (Any 1 × 2) (2)

[25]

TOTAL: 100

Copyright reserved